



## **Third Year Engineering**

### **21BTCS404 - USER CENTRED DESIGN**

#### **Class - T.Y. CORE(SEM-V)**

#### **Unit - IV** **Design Issues**

**AY 2023-2024      SEM-V**





## **Unit-IV**      **Syllabus**

Introduction to UX design process and case study,  
Understanding Information Architecture, Overview of tools.

**Exercise** - Open and closed card sorting technique -  
Creating information architecture for a system  
Understanding navigation models based on information  
architecture, High level concept  
sketches/wireframes.

**Exercise** - Creating low fidelity concept sketches for critical  
tasks of system/problems.

Quality of Service, Balancing Function and Fashion, User  
Documentation and Online Help,  
Information Search, Information Visualization.

# What is UI? What is UX?



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User  
Research

UI - User Interface  
GUI- Graphical User Interface

UX - User  
Experience



## Design

- the screen
- the elements
  - the color
- the platform



- The User
- Takes into consideration things outside the screen
- The environment, what the user might be doing



## Science & Psychology

- The study of Users, how they are using & experiencing the interface
- Experiments & Testing
- Questionnaires



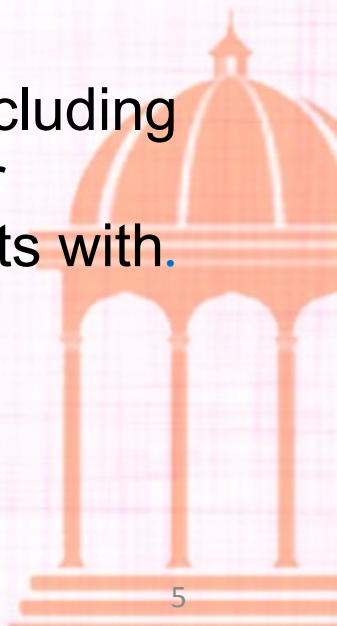
## What is UX Design?

- Products that provide great user experience are thus designed with not only the product's consumption or use in mind but also the entire process of acquiring, owning and even troubleshooting it.
- UX designers don't just focus on creating products that are usable, they concentrate on other aspects of the user experience, such as pleasure, efficiency and fun.
- is focused on the overall experience a user has while interacting with a product or service. It includes usability, functionality, and ensuring that the user's journey is seamless and efficient.
- A good user experience is one that meets a particular user's needs in the specific context where he or she uses the product.



## What is UI Design?

- **UI (User Interface) Design** focuses on designing visually appealing and intuitive interfaces for users to interact with digital products, such as websites, mobile apps, or software. The goal of UI design is to create a seamless and aesthetically pleasing experience that allows users to easily navigate and use a product. UI design is a crucial part of the overall user experience (UX).
- is concerned with the look and feel of the product, including visual elements like buttons, icons, typography, color schemes, and the overall layout that the user interacts with.





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Department of Computer Science & Engineering

## UI Vs UX Design



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| Aspect                        | UX Design                                                                                                           | UI Design                                                                                                                 |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Look vs. Feel</b>          | UX Design focuses on creating a feel and experience that addresses emotions and satisfaction                        | UI Design concentrates on visual aesthetics and elements with the aim of achieving visual appeal.                         |
| <b>Design vs. Prototyping</b> | In UX Design, wireframes and prototypes are created to define user flow and functionality                           | UI Design Finalizing visual designs for user engagement often involves working with mockups and graphics.                 |
| <b>High Level vs. Details</b> | The focus of UX Design is on ensuring a seamless and meaningful user journey by emphasizing the holistic experience | UI Design focuses on individual elements and interactions by perfecting details for aesthetics as well, as functionality. |



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## UI Vs UX Design

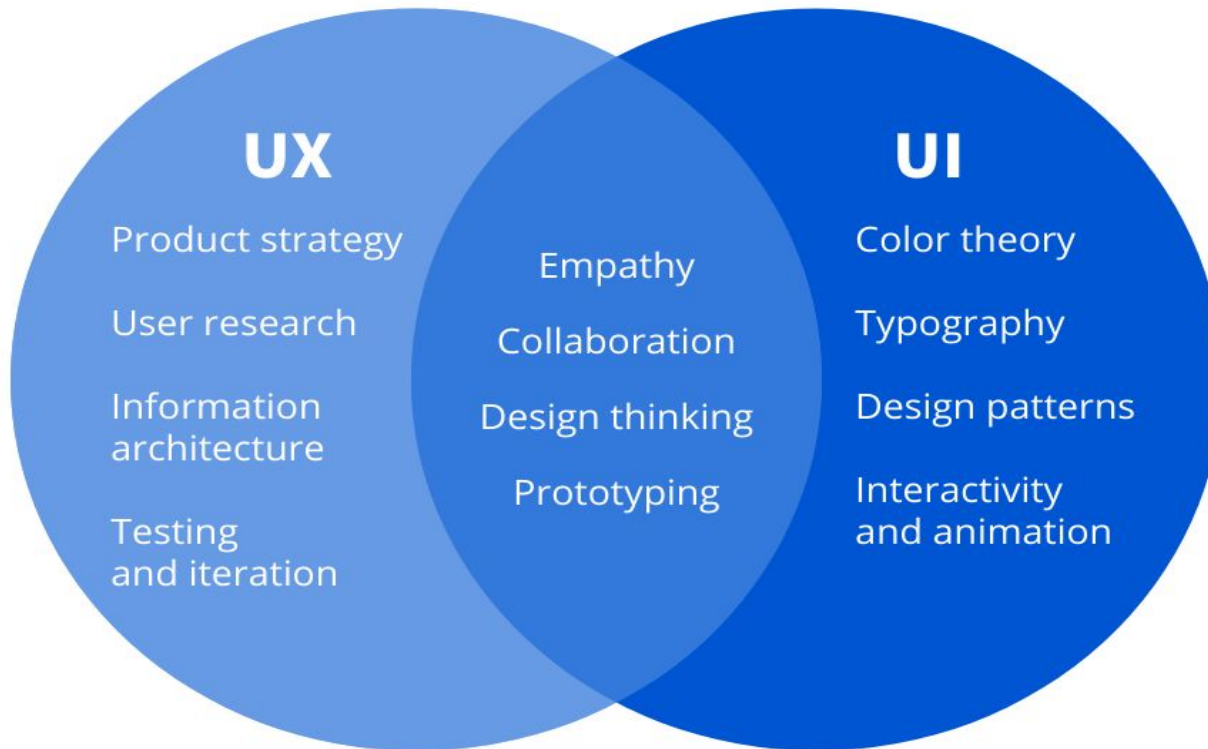


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|                                             |                                                                                                                       |                                                                                                                         |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Research and Implementation</b>          | Research involves conducting user analysis, testing, and developing strategies.                                       | Implementation on the hand focuses on putting design choices into action such, as selecting fonts, colors, and layouts. |
| <b>User-centered or Interface-centered.</b> | A user-centered approach prioritizes understanding user needs, behaviors, and emotions throughout the design process. | On the other hand, an interface-centered approach focuses on creating visually appealing and user-friendly interfaces.  |
| <b>Problem-Solving vs. Visual Appeal</b>    | Problem-solving aims to address user issues and alleviate their pain points                                           | Visual appeal focuses more on creating pleasing designs.                                                                |



## Relationship Between UX and UI



# UX & UI working together

## CONDUCT SCREENING AT AIRPORT ENTRANCES

A revolutionary long-term process change could be the dispersion of screening systems to elongated vestibules at airport entrances.

## LIMIT QUEUING SPACES

**Give passengers sole responsibility for tagging and dropping their own bags at check-in.**

## RETHINK SEATING IN BOARDING AREAS

We can look at new seating configurations that will allow for greater social distancing, and consider other flexible seating.

## PRE-MAP THE PASSENGER EXPERIENCE

Using technology, airports would assign passengers a dedicated time slot to pass through the security checkpoint.

**ACCELERATE  
BIOMETRIC SCREENING**

More airports can implement biometric screening processes that some locations have already begun testing.

## DECENTRALIZE PASSENGER PROCESSING

Spreading out passenger processing will be a key health and wellness strategy that airports can adopt in the near-term.

## **Key Stages in the Airport Journey**

- 1. Traveling to the Airport:** This stage involves planning and logistics, where users often seek real-time information about traffic, parking availability, and terminal navigation.
- 2. Arrival at the Airport:** Upon arrival, passengers face challenges related to check-in processes, such as locating check-in counters and understanding baggage drop-off procedures. Real-time updates about queue lengths and check-in locations can alleviate stress.
- 3. Security Screening:** Long wait times and unclear signage can heighten anxiety during security checks. Improvements in communication regarding expected wait times and clearer signage could enhance this experience.
- 4. Departure Lounge:** Once past security, passengers often struggle with visibility of amenities like dining options and restrooms. Many prefer to stay near their boarding gate, which can limit their exploration of available services. Providing clear visual cues and information about nearby amenities could encourage better utilization of facilities.
- 5. Boarding Process:** Effective communication about boarding times, gate changes, and potential delays is crucial. Passengers benefit from notifications about boarding issues before arriving at the gate to avoid last-minute rushes.

## **Common Pain Points**

- **Information Overload or Lack of Clarity:** Passengers frequently report confusion due to excessive or unclear information, especially regarding health checks or security requirements.
- **Physical Layout Issues:** Airports often face challenges with their physical layouts that hinder passenger flow. For example, split-level security operations can complicate navigation for travellers with mobility issues.
- **Technological Integration:** The integration of touchless technologies has become essential post-pandemic to enhance safety and streamline processes. Innovations like facial recognition for check-ins and touchless kiosks can significantly improve efficiency.

## Enhancing User Experience

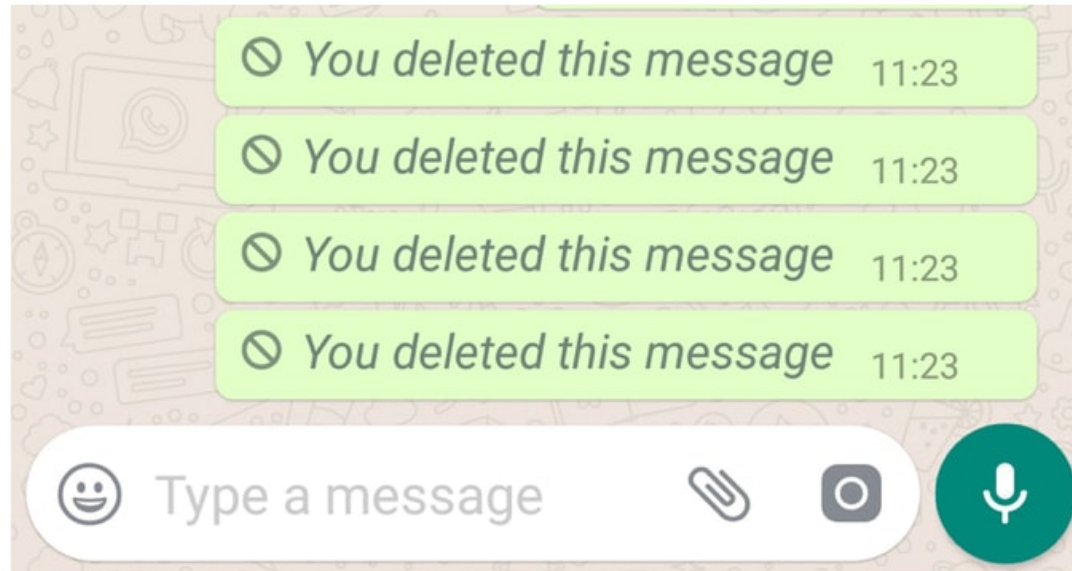
To improve the UX at airports, several strategies can be implemented:

- **Real-Time Updates:** Airports can leverage technology to provide passengers with real-time updates on flight statuses, queue lengths, and available amenities through mobile apps or digital signage.
- **Personalized Services:** Utilizing data analytics can help tailor services to individual passenger needs, such as personalized notifications about boarding gates or available shops based on their preferences.
- **Streamlined Processes:** Implementing self-service options for check-in and baggage drop can reduce wait times and empower passengers to manage their journeys more efficiently.
- **Holistic Design Approach:** Considering the entire passenger journey—from arrival to departure—can help identify critical touchpoints where improvements can be made to enhance satisfaction.

# Bad UX & UI can be deadly



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**Why it's bad UX:** Informing the recipient that the sender has deleted a message somewhat defeats the purpose of deleting it in the first place. In fact, this looks way more suspicious and is likely to prompt an awkward “why did you delete the message?” type of response.



# Bad UX & UI can be deadly



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Designs that are confusing, frustrating and just plain bad

Sometimes the best way to illustrate the value of design is through examples of bad design. And sometimes those examples happen in the real world.

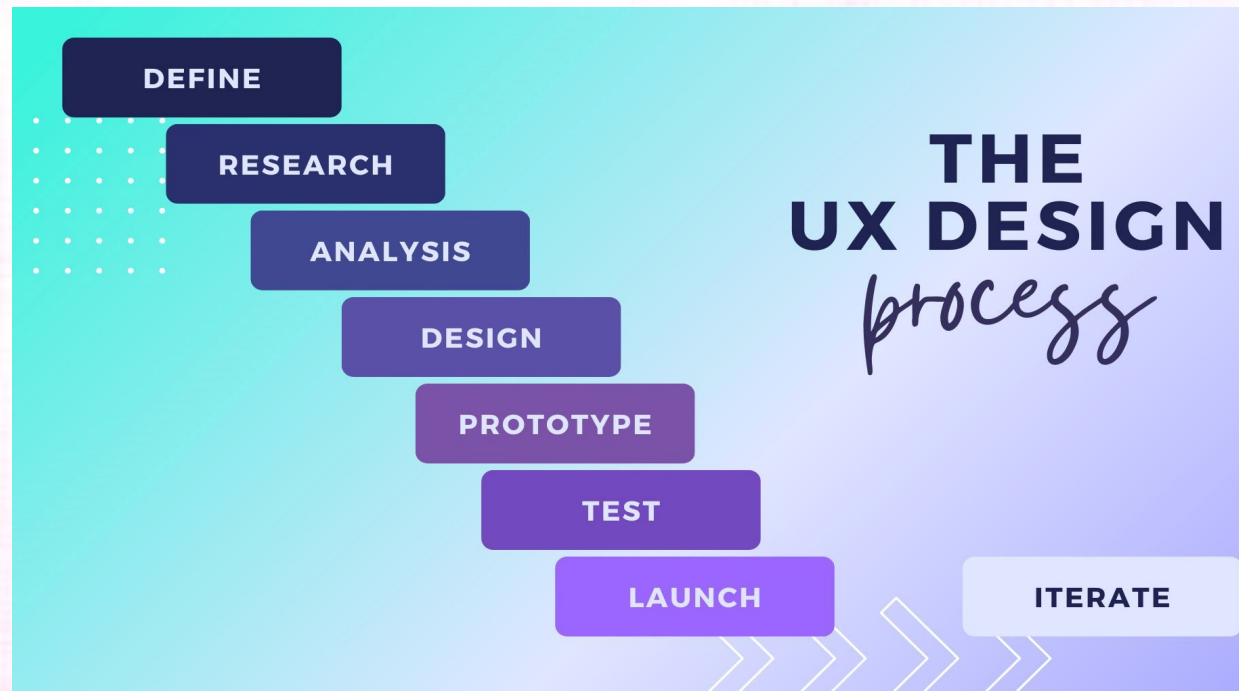
Good design often appears invisible—it doesn't get in the way of the user. But the designs in this post do the exact opposite. They needlessly confuse users, they assume users already have pre-existing knowledge and sometimes they are just plain thoughtless.



# What is a UX Design Process?

A UX design process is a series of steps that designers take to turn an idea into a user-friendly interface. The steps may vary depending on the project, but they typically include:

- Step 1: Define
- Step 2: Research
- Step 3: Analysis & Planning
- Step 4: Design
- Step 5: Prototyping
- Step 6: Testing
- Step 7: Launch
- Step 8: Iteration



# Why is the UX Design Process Important?



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The UX design process is important because it helps designers create interfaces that meet the needs of their users. By following a well-defined process, designers can be sure that they are covering all of the necessary bases and creating an interface that will be user-friendly and effective.



There are many different ways to approach the UX design process, but most designers follow these similar steps...

## **Step 1: Define**

In the first phase, it's necessary to determine exactly what needs to be created, and why. Why does this product need to exist? Who are you creating this for? What business problems will this solve? This is a conversation (or set of conversations) that often take place in the form of stakeholder meetings, where the product designers create a foundational approach that aligns with a high-level business strategy.

Out of this initial kick-off meeting, you'll likely have a set of specs to work from, as well as a low-fi concept sketch. This gives you a basic outline that can then be used in the next step.



## Step 2: Research

- In this phase, designers conduct research to gain a deep understanding of their users and their needs.
- This research helps them create empathy for their users and understand what they need from the product or service.
- Both user research and market research are necessary components of this phase.
- User research can take many forms, such as interviews, surveys, focus groups, and ethnographic studies. Market research looks at things like industry trends and competitive analysis.
- There are a few different ways to go about conducting user research. One popular method is customer journey mapping, which allows you to see the steps your users take as they interact with your product. Another common technique is usability testing, which gives you direct feedback from users on what works well and what needs improvement.

## Step 3: Analysis & Planning

- In the planning phase, designers take all of the information they gathered in the research stage and start to plan out how they'll meet those needs.
- They develop user personas, **user stories**, **wireframes**, and other high-level plans during this phase.
- This is also the time when designers start to think about how the product will be built and what technologies will be necessary.
- They develop a roadmap for the project and start to establish Milestones.

## Step 4: Design

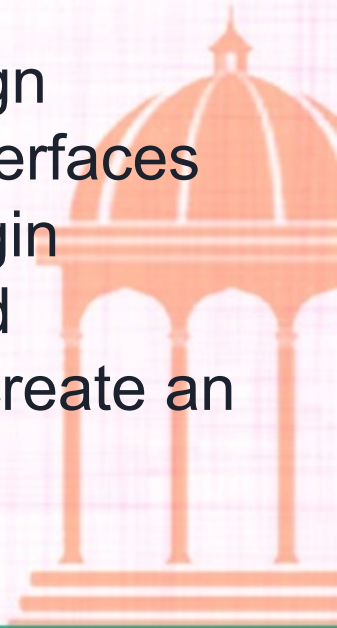
- Once you have a good understanding of your users and a plan to move forward, it's time to start sketching out some ideas for how they'll interact with your interface.
- This is where you'll start to think about things like the overall layout, navigation, and specific elements on each page.
- The key here is to think about the user experience first and foremost. How will they interact with your interface? What kind of information do they need to be able to find easily? By answering these questions, you can start to form a clear picture of what your interface should look like.

## **More about Designing Phase.....**

This design phase typically includes both UX and UI aspects, since you'll work on things like:

- Information architecture
- Navigation
- Layout
- Usability and accessibility
- Microcopy

There will also be a shift between the UX and UI design phases, where you turn your wireframes and low-fi interfaces into something more polished. This is where you'll begin working on things like color schemes, typography, and iconography. All of these elements come together to create an aesthetically pleasing and easy-to-use interface.





## **Step 5: Prototyping**

Once you have a working UI, it's time to turn it into a working prototype. Prototyping allows you to present a more realistic experience for your usability testing, which can then deliver more accurate feedback and insight into "what's working" and "what's not working".





## Step 6: Testing

- Before launch, it's important to test the interface with real users. **Usability testing** helps identify any areas that need improvement before the final product goes live, and delivers this feedback from the user's point of view.
- The more information you receive from your testing, the easier it will be to identify exactly what needs to be revised before you ship the product.



## Step 7: Launch

Once the testing is complete and all of the necessary changes have been made to your high fidelity user interface, the product is ready to be handed over to the development team for implementation.

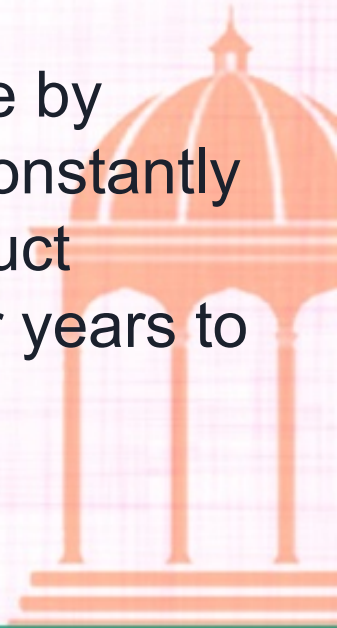


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## Step 8: Iteration

After launch, the product—whether that's a website, app, or other digital product—is not done. The design process is an ongoing cycle that's meant to be repeated as users interact with and provide feedback on the product.

The goal is to continuously improve the user experience by making small changes and refinements over time. By constantly iterating on your design, you can ensure that your product remains usable and relevant to your target audience for years to come.





# Understanding Information Architecture

Think of a time when you had to click through more than five links on a website in search of information, only to come up empty-handed. This is the outcome of weak information architecture - a frustrating experience for the user. No one wants that.

To avoid losing users, designers organize content into easier-to-consume systems. The science of structuring information, features, and functionalities in digital products is information architecture (IA). IA is a fundamental concept that UX designers should master before they begin any visual



# What Is Information Architecture



1. Information architecture is the science of organizing, prioritizing, and labeling content to structure digital products. Visualize it as a blueprint of the product's infrastructure and features.
2. A UX designer uses information architecture to deliver those to the user in a way that's clear to understand and simple to navigate.

Think of your favorite website or app. Now try naming information types, features, structures, and content that go into the platform. If you had to draw how all of those elements work together, where would you start? That's **the job of an information architect**.

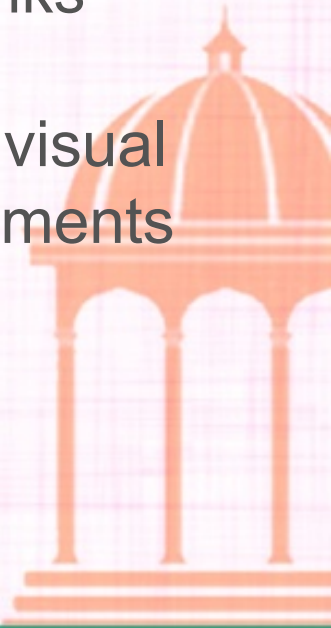


# The Role of Information Architecture

Information architecture falls under the responsibilities of a UX designer.

A **UX designer's main focus is to create a user-centered experience.** To achieve this, the designer starts by creating a skeleton to navigate a web page. This structure is known as information architecture. It sets the relationship and links between the different elements.

Think of any architecture design. This blueprint is the visual deciding where to place stairs or doors to connect elements of a building.



As the name suggests, the organization system aims to organize information. There are many ways you can divide and classify information. An organization system gives you a framework to follow so that your product makes sense to your users.

The organization system can have one of these structures: hierarchical, sequential or matrix.

1. The hierarchical structure uses visual indicators to assign importance to the different elements of a design. Using color, size, contrast, shape, alignment, and other visual clues, you can make some elements more prominent than others. This structure aims to direct the user's attention to more important elements and to stimulate the brain's next interest point.

2. The sequential structure draws out the possible journey a user could follow. It's a sequence of actions a user takes to reach a goal. This structure is very common in retail and food ordering applications.

One popular example is the process of ordering a pizza.

**When ordering a pizza online, you follow a set process. You first decide your size of pizza, crust, cheese, sauce, toppings, then you proceed to payment.** The platform takes you on this intended journey for you to complete your purchase.



### 3. Matrix Structure

The matrix structure is a bit more complex, as the navigation is in the hands of the user. A phone interface is a perfect example of a matrix structure. In this case, the architecture is customizable - in fact, the more customizable the homepage, the more time the user will spend interacting with it.

The user has choices for organizing the information, and can use different variables to sort and arrange it. For example, the user can decide to view information by date or type. This changes not only the information structure but the interface of the platform.

This structure is used when there are multiple paths that a user can take to get to what they need.



# Labelling System

Labels work like containers - they assemble a large amount of information under one tag. In design, these tags can be represented by headers as an example. A header can signal that there is more content and information inside.

This helps sort data on the back end, but also presents a great experience for the user.

## Navigational System

The navigational system determines how users travel through your content. This system determines how the pages interact with each other and how accessible the content is. A good navigation helps the user explore the website without getting lost.

These navigational systems can be:

- Global or site-wide, like navigation bars on a homepage with links to other sections of the same website
- Local, which helps the user navigate through different sections of the same page
- Contextual, which can vary but an example of which is moving to a different website

# Searching System

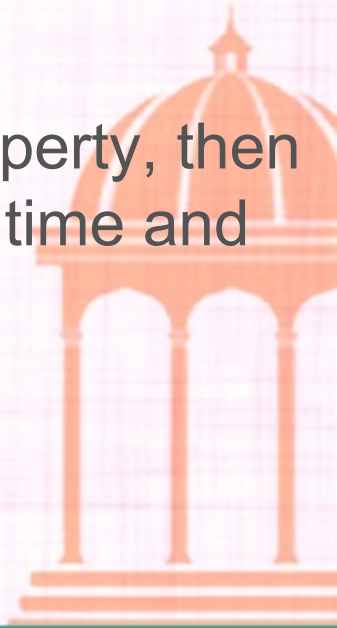


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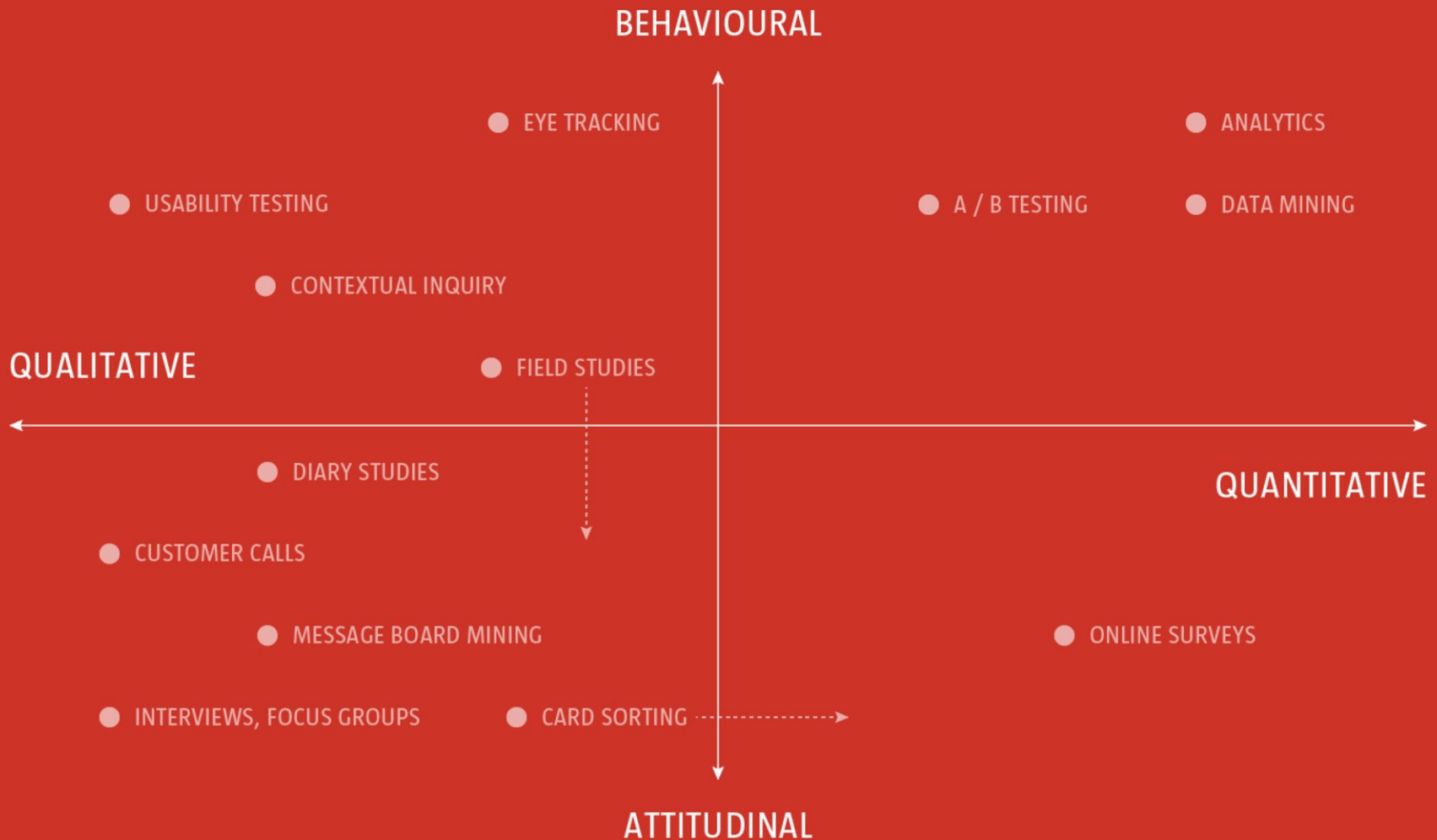
A searching system helps users find specific content or related content using keywords. This system relies on a keyword-driven search engine or a set of filters to help discover content.

Real estate websites are a good example for this type of system, as finding the right apartment on a website relies heavily on a searching system. Not all listed properties are on one main page - the list would go on forever, and you would never see any of your interests.

That's why architects gather information from each property, then create filters for pricing and features which save users time and headaches.



# UX research tools:





We can think of the research landscape as one with two axes:

- Qualitative and Quantitative; and
- Behavioral and Attitudinal
  - On one axis we have qualitative approaches, exploring opinions, quantitative approaches, and testing opinions by exploring data.
  - On the second axis we have behavioral approaches, observing users, watching what they do; and attitudinal approaches, listening to users, exploring their attitudes and opinions.



- **Qualitative** research tools focus on understanding *why* users behave in a certain way by exploring their attitudes, emotions, and motivations. This involves collecting non-numerical data like interviews, observations, or open-ended responses.
- **Quantitative** research tools aim to measure *what* users do and how often they do it, providing statistical data through methods like surveys, A/B testing, or analytics.  
Furthermore, UX research can gather:
- **Behavioral** data, which shows what users *do* when interacting with a product.
- **Attitudinal** data, which reveals what users *say* they think or feel about a product or experience.

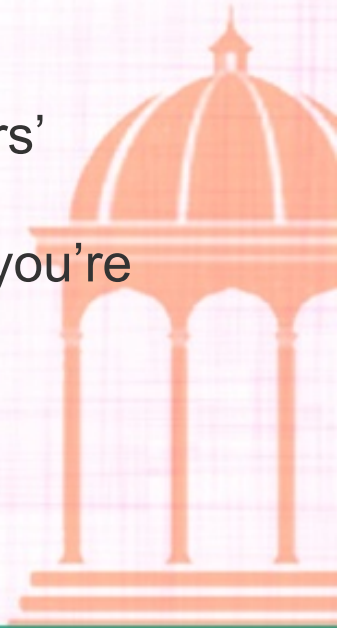


# QUALITATIVE RESEARCH

Qualitative research is primarily exploratory research, undertaken to establish users' underlying motivations and needs. It's useful to help you shape your thinking and to establish ideas, which can then be built and tested using quantitative methods.

## INTERVIEWS

Interviews are a great way to really get to the heart of your users' needs. To facilitate interviews effectively requires a degree of empathy, some social skills, and a sense of self-awareness. If you're not this person, find someone on your team who is.





Broadly speaking interviews fall into one of two categories:

- Structured Interviews:** Where the interviewer focuses on a series of structured questions, comparing the responses of the interviewee to other interviewees' responses.
- Semi-Structured Interviews:** Where the interviewer adopts a looser, more discussion-driven approach, letting the interview evolve a little more naturally.



# CONTEXTUAL INQUIRY



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A contextual inquiry is focused around four key principles:

- Context:** Interviews are conducted in the user's workplace, which affords an opportunity to experience typical working conditions, existing solutions and, equally, a user's frustrations.
- Partnership:** The researcher and user work together to understand the user's workflow and the tools they use.
- Interpretation:** By sharing the researcher's observations and insights with the user, there is an opportunity for the user to clarify or extend the researcher's findings.
- Focus:** The researcher is able to guide the user's interactions towards aspects that are relevant to the specific project's scope.



# CARD SORTING

- **Card sorting** is a UX research method used to understand how users categorize and organize information. It helps designers create intuitive information architectures (IA) for websites, apps, or other digital products by revealing how users expect content to be grouped or labeled.
- This method is especially useful for designing navigation menus, structuring websites, or organizing products in e-commerce platforms.

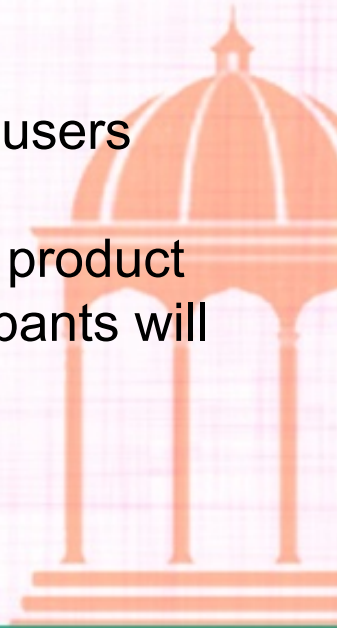


# CARD SORTING

- **Open Card Sorting:**
  - Participants are given a set of cards (each representing a piece of content, feature, or item) and asked to group them into categories. They can create their own categories and name them.
- **Use case:** When you're looking for insights into how users think content should be organized without predefined constraints.
  - **Example:** Asking users to categorize different features of a mobile banking app, allowing them to create and name groups like "Transactions," "Account Management," and "Support."

Scenario: Designing a new e-commerce website.

- **Goal:** You want to organize the product categories based on how users think.
- **Process:** You give participants a set of cards, each labeled with a product type, and ask them to group the cards into categories. The participants will create their own category names based on their understanding.



# CARD SORTING

## Cards Provided:

- T-Shirts
- Laptops
- Headphones
- Sneakers
- Jackets
- Mobile Phones
- Watches
- Sunglasses

## Example Groupings by Participants:

### •Participant A:

- **Electronics:** Laptops, Mobile Phones, Headphones
- **Clothing:** T-Shirts, Jackets
- **Accessories:** Watches, Sunglasses
- **Shoes:** Sneakers

### Participant B:

- Tech Gadgets:** Laptops, Mobile Phones
- Fashion:** T-Shirts, Jackets, Sneakers, Watches, Sunglasses
- Audio:** Headphones



# CARD SORTING

- **Closed Card Sorting:**
- Participants are given a set of cards and a predefined set of categories. They must place each card into one of the categories.  
**Scenario:** Organizing a knowledge base for a customer support website.
- **Goal:** You already have predefined categories and want to see how users place specific topics under these categories.
- **Process:** You provide participants with a set of cards and predefined categories. Their task is to place each card in the most suitable category.



# CARD SORTING

Categories Provided:

- Account Issues
- Technical Support
- Billing
- Shipping

Cards Provided:

- Password Reset
- Credit Card Update
- Delivery Time
- Software Installation
- Tracking a Package
- Forgot Username
- Refund Request
- Troubleshooting



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Example Sorting by a Participant:

•**Account Issues:**

Password Reset,  
Forgot Username

•**Technical Support:**

Software Installation,  
Troubleshooting

•**Billing:** Credit Card  
Update, Refund  
Request

•**Shipping:** Delivery  
Time, Tracking a  
Package





- **Focus Groups:**

- Group discussions where multiple users provide feedback on a product, feature, or concept.
- Helps gather diverse perspectives and attitudes, but group dynamics may affect individual opinions.

**Example:** Gathering a group of 8 users to discuss their experiences with a new e-commerce website.

- **Contextual Inquiry / Field Studies:**

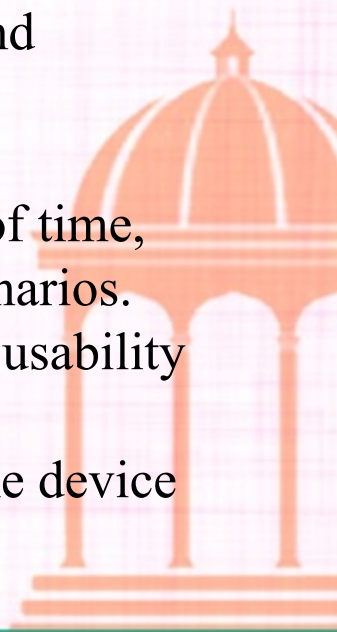
- Researchers observe users in their natural environment while they use a product or perform tasks.
- Provides rich insights into how users behave in real-world contexts, highlighting challenges and opportunities that may not be apparent in controlled settings.

**Example:** Observing employees using software in an office to understand workflow issues.

- **Diary Studies:**

- Participants record their activities, thoughts, and feelings over a period of time, providing insight into their experiences with a product in real-world scenarios.
- Captures long-term behaviors and patterns that may not emerge in short usability tests.

**Example:** Asking users to document their interactions with a smart home device over a week.



- **Quantitative research** is a research method that focuses on collecting and analyzing numerical data to identify patterns, relationships, and trends.
- It aims to quantify the problem by generating statistical data that can be used to make predictions, draw generalizations, or establish facts.
- Quantitative research often answers questions like *how many*, *how much*, or *how often*, and it is typically used when researchers want to test hypotheses or measure the extent of a particular phenomenon.



# SURVEYS AND QUESTIONNAIRES

- Surveys and questionnaires are a powerful tool for gathering a higher volume of opinions.
- Structured questions that generate numerical data, often using closed-ended questions (e.g., yes/no, multiple choice, or rating scales).
- Surveys can be administered online, via phone, or in person to collect data from a large sample.
- Example: Conducting a customer satisfaction survey with a 5-point Likert scale (1 = Very Dissatisfied, 5 = Very Satisfied) to quantify overall satisfaction levels.



- Collection and analysis of user behavior data on digital platforms (e.g., websites, apps) through tools like Google Analytics.
- Measures metrics such as page views, click-through rates, time spent on page, bounce rates, and more.
- **Example:** Analyzing user traffic and conversion data to determine the effectiveness of an online marketing campaign of e-commerce website.
- Promoting a new product line. You want to understand how well the campaign is driving traffic and converting visitors into customers.



We're fortunate now to have considerable quantities of data at our fingertips. Tools like **Google Analytics** — the most widely used web analytics service — let you measure website traffic and generate reports quickly and easily. Analytics, although occasionally a little overwhelming, are great for testing your assumptions.

Drawn from data, **analytics can be a persuasive tool** when working with business executives who, more often than not, “Want to see things in black and white.” Having easy access to numbers of unique visitors, page views, pages per visit and other metrics allows you to test your thinking once you've implemented your design after your qualitative research phase.



- A form of controlled experimentation where two or more variants of a web page, app feature, or product are tested to determine which performs better based on user actions.
- Commonly used in UX design to compare different versions of a website or interface.
- **Example:** Comparing two different versions of a homepage (A and B) to see which one results in a higher conversion rate.



A/B testing is another useful tool to test out multiple ideas to see which ones work best. Essentially a controlled experiment with two variants, A and B, this form of research allows you to test different designs effectiveness against each other.

A/B testing works well when you have large sample sizes. With a lot of traffic to a website, or a large mailing list, you can be more confident that your findings are backed by a substantial quantity of data.

A/B testing, also known as split testing, is a method of comparing two versions of a webpage, email, or ad to see which performs better based on a particular metric, like conversion rate, click-through rate (CTR), or user engagement.



| Type       | Qualitative                                                                                                                                                                                                                                                                                                                                                                                                                      | Quantitative                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Behavioral | <b>Qualitative Behavioral Tools</b><br>- Directly observe what users <i>do</i> and gather non-numerical data.                                                                                                                                                                                                                                                                                                                    | <b>Quantitative Behavioral Tools</b><br>- Capture user actions and interactions in a measurable format                                                                                                                                                                                                                                                                                                               |
|            | <ul style="list-style-type: none"> <li>- <b>User Interviews:</b> In-depth conversations to uncover user behavior and motivations.</li> <li>- <b>Field Studies:</b> Observing users in their natural environment to see how they interact with a product in real-life settings.</li> <li>- <b>Usability Testing:</b> Directly observing users as they try to complete tasks (using tools like Lookback, UsabilityHub).</li> </ul> | <ul style="list-style-type: none"> <li>- <b>Analytics (e.g., Google Analytics):</b> Tracks user interactions such as clicks, time on page, or conversion rates.</li> <li>- <b>Heatmaps (e.g., Hotjar, Crazy Egg):</b> Shows where users click and scroll.</li> <li>- <b>A/B Testing (e.g., Optimizely, Google Optimize):</b> Compares different versions of a product to determine which performs better.</li> </ul> |
|            |                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Type        | Qualitative                                                                                                                                                                                                                                                                                                                                   | Quantitative                                                                                                                                                                                                                                                                                                                                            |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Attitudinal | <b>Qualitative Attitudinal Tools</b><br>- Focuses on what users say they think or feel.                                                                                                                                                                                                                                                       | <b>Quantitative Attitudinal Tools</b><br>- Measures what users say they think or feel, often through surveys.                                                                                                                                                                                                                                           |
|             | <ul style="list-style-type: none"> <li>- <b>User Interviews:</b> Open-ended conversations to explore user perceptions.</li> <li>- <b>Focus Groups:</b> Facilitated group discussions to collect opinions about a product.</li> <li>- <b>Card Sorting (e.g., Optimal Workshop):</b> Understanding how users categorize information.</li> </ul> | <ul style="list-style-type: none"> <li>- <b>Surveys (e.g., SurveyMonkey, Typeform):</b> Structured questions to gather user feedback, often yielding a large data set of opinions.</li> <li>- <b>Net Promoter Score (NPS):</b> Measures user satisfaction and loyalty with a simple survey question ("How likely are you to recommend...?").</li> </ul> |
|             |                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                         |



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# Thank You

